

Use-Case Flow Specification

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Problem no: 8

Section: H

Use Case: UC-02 — Place Hold Request

Primary Actor: Student Member

Secondary Actor: none (Notification System is invoked in a later use case, UC-08 Process Book Return)

Related Requirement: FR-001, FR-002

Preconditions

1. The Student Member is authenticated and logged in to the library portal.
2. The requested title exists in the ISBN catalog and all physical copies are currently on loan (zero copies available).

Postconditions (on success)

1. A new hold record exists for the member, with a FIFO queue position assigned for that title.
2. The member's active loan-and-hold count is incremented by one.

Main Success Scenario

1. Student Member searches for a title by ISBN or keyword (UC-01 Search ISBN Catalog).
2. System displays the search results, showing the title's availability status as “All Copies Loaned.”
3. Student Member selects “Place Hold” on the fully-loaned title.
4. System executes «include» Verify Borrowing Eligibility (UC-06), checking the member's current active loan-and-hold count against the maximum allowed (5 items).
5. System confirms the member is under the limit and eligible to place a hold.
6. System creates a hold record and assigns the member the next FIFO position in the title's waitlist.
7. System displays the confirmed queue position and an estimated wait time to the member.
8. System logs the hold event. Use case ends successfully.

Alternate Flow — 5a. Member Already at Maximum Hold Limit

- 5a1. During Verify Borrowing Eligibility (step 4), the system determines the member's active loan-and-hold count is already at or above the maximum allowed (5 items).
- 5a2. System displays a “Hold limit reached — cannot place further reservations” message, stating the limit and the member's current count.
- 5a3. System declines to create the hold record and offers the member a link to view or cancel an existing hold.
- 5a4. Use case ends without a hold being created (matches the FR-001 “Fail” acceptance criterion).